

**CCC Tour Guide: A 3D Virtual Reality Desktop Application for Visualizing and Interacting with
Virtual Elements with Binary Decision Tree Algorithm**

A Thesis Presented to the
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ABSTRACT

With the on-going pandemic, students nowadays rarely go to the school in order to follow strict health protocols to prevent the spread of the virus. As computer science students, the researchers came up with the idea to develop a 3D Virtual Reality Desktop Application to tour the institution's visitors and senior high school students from different institutions and to promote the institution's facilities, structural and architectural design to new-comers, especially, new enrollees and visitors to help them get the inspiration that they need to study in the institution once the pandemic is over. The application can be used to roam the institution virtually without the need to travel to the physical location of the institution. Furthermore, it can be used to interact with the virtual elements that has been created for the users to enlighten some of the information about the institution. Moreover, the application is assisted by the Binary Decision Tree algorithm. The algorithm produces an outcome of choices based on the architectural design of the environment. Also, virtual elements are also available in the application to show some information about the institution. Furthermore, a Developmental research approach is used. Since the study is experimental research, the researchers have employed the Waterfall model as a System Development Life Cycle model. An online survey was also conducted to satisfy the requirements of the panels.

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CHAPTER I

INTRODUCTION

Project Context

Nowadays, technology is evolving rapidly at a pace that is faster than before. Every year, new technologies are being developed simultaneously because of the enormous commissions offered if quotas are met. Technological sales is one of the most financially profitable career options in businesses. A technology that is so popular and being used almost everyday is Virtual Reality or VR. Virtual reality (VR) is a virtual experience that can be both comparable and different to the actual world. Virtual reality has a variety of uses, including entertainment (such as video games), education (such as medical or military training), business (e.g. virtual meetings), tour guide (e.g. Google Maps), and many other uses.

From 1970 through 1990, the virtual reality industry primarily produced VR systems for medical, flight simulation, vehicle design, and military training. But in the mid-1990s, Virtual reality headsets for consumers were first introduced by video game firms (The Board of Trustees of the University of Illinois, 1995). Oculus (Rift), HTC (Vive), and Sony (PlayStation VR) all introduced next-generation commercial tethered headsets in the 2010s, igniting a new wave of application development (F. Rohr, 2015). Nowadays, Virtual Reality is often used for tour guiding (e.g. Google Maps) and travelling purposes. Such applications are helpful for people who do not travel that much and are not familiar with many places. But then, a pandemic hit many different countries that interrupted, not just many transportation services, but also many other businesses and also the education system.

With the on-going pandemic, students nowadays rarely go to the school in order to follow strict health protocols to prevent the spread of the virus. The Commission on Higher Education, also known as CHED, which is the Philippine government's agency that is in-charge with the promotion of relevant and quality higher education, announced, in May 2021, that flexible learning will become the standard in the future. There are no plans to return to traditional, overcrowded face-to-face classes. Flexible learning will be continued in

School Year 2021 and later, according to the commission. Hence, making technologies more utilized and implemented.

The main objective of this study is to implement a new approach to tour new students and visitors through virtual reality that does not require a physical approach to visit the institution. This study also aims to aid the need of tour assistants for visitors and students. With the use of virtual reality technology, this application can help to, not just improve, but also elaborate even more some of the basic information about our institution while complying with the health protocols of the government.

Theoretical Framework

This study is anchored on Decision Theory. As K. Steele (2020) stated “Decision theory is concerned with the reasoning underlying an agent’s or a user’s choices”. Also, when the implications of a choice are unknown, Decision Theory gives a reasonable framework for selecting amongst various courses of action. Utility theory and the inductive application of probability theory serve as the basis (D. W. North, 1968).

This study is also supported by Utility Theory. As B. J. Cohen (1996) stated “Expected utility theory is felt by its proponents to be a normative theory of decision making under uncertainty”. The foundation of the theory is a set of simple axioms that are assumed to represent rules that any reasonable individual would follow. It can be proved that if these axioms are followed, each conceivable outcome can be assigned a numerical number known as utility, with the desired course of action being the one with the highest expected utility. Moreover, the expected utility of an action is a weighted average of the utilities of each of its possible outcomes, with the utility of a result indicating how much it is favored over the alternatives. Each outcome's utility is weighted based on the likelihood that the act will result in that outcome. (R.A. Briggs, 2019)

This study is also supported by Normative Decision Theory. As stated by Edwards (1954), “Normative theories identify "good" decisions as those that are most likely to result in the intended outcomes for the decision maker”. Moreover, according to R. Sanrainte (2016), the most optimal decision for a given scenario is modelled by normative decision theory. A perfectly rational actor is postulated in normative theory. Normative decisions are always made with the goal of achieving the highest expected value. A fully rational agent is capable of precisely calculating the highest expected value.

Conceptual Framework

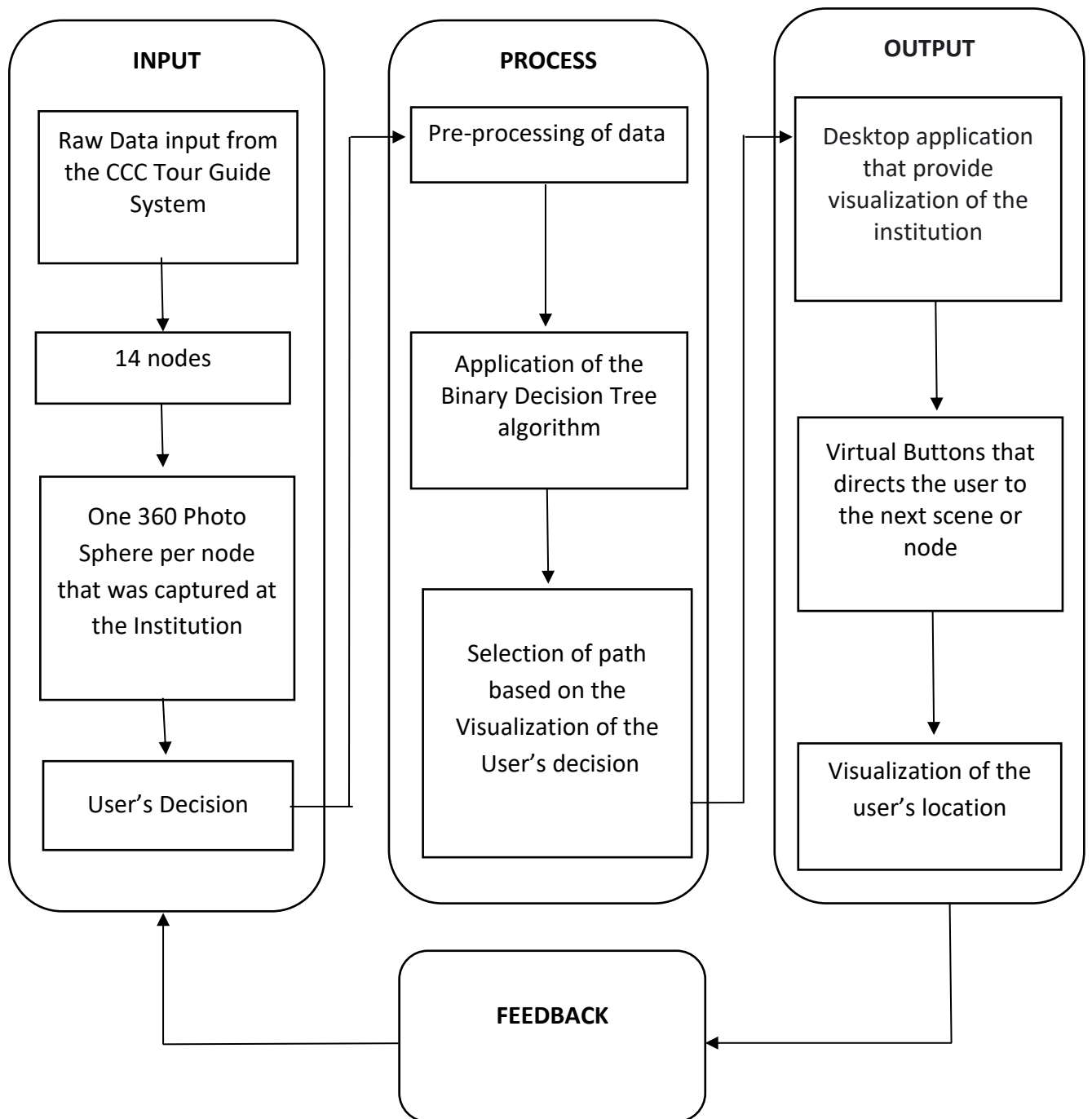


Figure 1: Conceptual Framework

The input consists of raw data input from the CCC Tour Guide System. The system consists of fifteen nodes that represent fourteen 360 Photo Spheres. The 360 Photo Spheres were captured inside and outside of the institution. The User's decision will change the outcome of the process and output.

The process consists of the pre-processing of the inputted data to be processed by the algorithm. Then, the selection of paths will be based on the user's decision.

The output will be a desktop application that provides visualization of the institution. The desktop application contains the Virtual Buttons that directs the user to the next scene or node. The output also contains the visualization of the user's location.

Objectives

General Objectives

To create a virtual reality desktop application that is based on City College of Calamba's real-life environment to help the users visualize and interact with the virtual elements in the system.

Specific Objectives

- To design and develop a 3D Virtual Reality Desktop Application that can provide a new interactive technology to be experienced by the students, faculty, and alumni.
- To promote the institution's facilities, structural and architectural design to new-comers especially new enrollees and visitors to help them get the inspiration that they need to study in the institution once the pandemic is over.
- To be able the institution to be visualized by the students, faculty, and alumni without having to actually be in the environment and to elaborate even more some of the basic information about the institution while complying with the health protocols of the government.
- To create a structure of choices based on the institution's interior architectural design that will be assisted by the Binary Decision Tree Algorithm to be used on the program.

Significance of the Study

This study was anticipated to contribute additional information to serve the following individuals and institutions.

To Senior High School students. To promote City College of Calamba to Senior High School undergraduates.

To Students. They may explore this study for their future references.

To Professors. This study will aid them to their discussions regarding related lessons. It will be easier for them to tackle related topics about this study.

To Future Researchers. This study will help the future researcher as they may use this study as their basis or a guide in formulating their own research design and will provide an additional source of information.

Generally, this research study will aid the City College of Calamba: students, professors, and the future researchers and the researcher itself.

Scope and Limitations

This study extends its function to users who only have Windows as the operating system to use the Virtual Reality Application. The application cannot be installed on android or IOS mobile devices. The application can only be used by students and alumni, senior high school students or visitors, and, lastly, staff and faculty of City College of Calamba.

This study focuses on designing, developing, and discussing how the application was made using the Waterfall model of System Development Life Cycle. The application does not need an internet connection. The developers will send a copy to the institution and they will manage who can have access to the application. The application does not contain a map of the institution. It also does not have user registration or login to control who will have the access of the application since the institution will choose who will receive the application. It does not require the use of Virtual Goggles since it is not affordable for everybody.

The application will only show what the actual look of the institution is and would not be modified for beautification purposes. The only requirement of this system to be used is a mouse and a monitor. The mouse can be used for travelling, for moving around the point of view, and zooming while the monitor will display the institution as a virtual reality. The application cannot search or identify where a room is located. The only purpose of this application is to view and explore freely the institution.

The proponents used Google Street View to develop the 360 Photo Spheres that is being used in the application and an android mobile device to capture the 360 Photo Spheres from the institution.

Definition of Terms

For purposes of clarity in the meaning of terms used, the following definition were presented:

- **Tour Guide** - a person who is hired to show tourists around tourist attractions.
- **Virtual tour** - a simulation of an existing location, usually composed of a sequence of videos or still images.
- **3D** - a three-dimensional representation
- **Virtual Reality** - the application of computer modeling and simulation to allow a human to interact with a three-dimensional (3-D) visual or other sensory world that is artificial
- **Virtual Elements** - represent temporary fields in applications.
- **Binary Decision Tree Algorithm** - a Boolean decision tree. Each branch leads to a new tree stump, which is called a node, from which it again splits in two. At every node there is a true or false question. The two branches coming out of the node correspond to the two possible answers (true or false) to the questions.
- **Decision Tree Algorithm** - belongs to the family of supervised learning algorithms. Unlike other supervised learning algorithms, the decision tree algorithm can be used for solving regression and classification problems too.
- **Decision Tree** - a decision support tool that uses a tree-like model of decisions and their possible consequences, including chance event outcomes, resource costs, and utility. It is one way to display an algorithm that only contains conditional control statements.
- **Binary** - consisting of two things or parts (e.g. left or right, true or false)
- **Axiom** - a statement or proposition which is regarded as being established, accepted, or self-evidently true.

- **Algorithm** - a process or set of rules to be followed in calculations or other problem-solving operations, especially by a computer.
- **Tree** - a widely used abstract data type that simulates a hierarchical tree structure, with a root value and subtrees of children with a parent node, represented as a set of linked nodes.
- **Virtual** - not physically existing as such but made by software to appear to do so.
- **Architectural Design** - is a term that focuses on structural components or aspects
- **Interior Architectural Design** - is the design of a new interior for a type of property that can be fixed, or the design of a building or shelter from the inside out
- **Classification Algorithm** - a function that weighs the input features so that the output separates one class into positive values and the other into negative values.
- **Nodes** - a basic unit of a data structure, such as a linked list or tree data structure. Nodes contain data and also may link to other nodes.
- **Boolean** - a binary variable, having two possible values called "true" and "false".

CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDY

This part provides the local and foreign related literature and studies to supplement and support the ideas of the researchers. This will provide the necessary information to direct and shed light to the problems at hand. The focus of this review is to take the relevant local and foreign literature and studies as well as to better understand the relationship of technologies.

Related Literature

Local Literature

The focus of this part is to take the relevant local literature as well as to better understand the correlation between technologies that is focused on this study.

As studied by Figueroa, R. B., et al. (2020). In this regard, the rising availability of low-cost technology for experiencing virtual reality (VR) apps, as well as the spread of 360° or virtual reality (VR) photo cameras, give a practical and virtual alternative. This is a description of a pilot project that used a virtual reality photo tour to replace ICU's typical physical campus tours. It presents a usability analysis and a preliminary draft measure of user experience derived from preliminary results such as the effect of spatial presence on usability, desire to visit, and variables related to interest and essential motivation obtained after conducting correlation analysis, t-test, and text analysis and obtaining preliminary results such as the effect of spatial presence on usability, desire to visit, and variables related to interest and motivation. The major findings indicate how a virtual reality campus tour may complement and enhance in-person campus visits when used effectively, while also allowing for more research into spatial presence and motivation in educational settings.

In conclusion of this local literature, according to preliminary data, the virtual tour's spatial presence increased participants' interest in the university and their desire to visit the campus. The tour was also an

effective tool for informing tourists about the university. The SUS scores indicate that the tour itself has to be enhanced, however because of the constraints of Story Sphere's beta version, it will be impossible to make the recommended and required changes. The news is not all bleak though, as the preliminary results of this study reveal that a virtual tour, made on a free platform and with minimal resources can be effective in complementing in-person tours. The outcomes of this study are supposed to enlighten future researchers about indicators that could indicate the utility of a virtual tour, such as usability, intent to visit, and perceived usefulness in learning about the university.

The above literature discussed that the virtual tour's spatial presence increased participants' interest in the university and their desire to visit the campus. The virtual tour was also an effective tool for informing tourists about the university.

According to Palad, et al. (2020). Based on such initial data, the decision tree algorithm success is due to alternative methods. The advantage of employing a decision tree classifier over other classification algorithms, as per some, is that the tree can be easily visualized, recognized, and interpreted because it has a tree-like structure. Other researchers have come to the same conclusion that the decision tree classifier is one of the most efficient and widely used classifiers.

In conclusion of this local literature, the initial study that prompted this work was inspired by the need to investigate data mining techniques in the field of cybercrime investigation in the Philippines. Using only 14,098 mostly Filipino words, the research study investigated the performance and prediction accuracy of three classifiers in distinct categories: J48 decision tree, Naive Bayes, and SMO. Furthermore, their findings show that the J48 classifier beat the other two classifiers, having the highest accuracy and lowest error rates. During their validation experiment, police investigators also preferred the J48 classifier. No research had been undertaken or concluded in the country using cybercrime data and data mining prior to this work, based on the authors' knowledge.

The above literature discussed that using only 14,098 mostly Filipino words, the research study investigated the performance and prediction accuracy of three classifiers in distinct categories: J48 decision tree, Naive Bayes, and SMO. Therefore, Decision trees were effective as a tool for prediction.

As indicated by Las Johansen, B. C. (2018). Decision trees are standard supervised learning algorithms, easy to understand and easy to use. Decision trees have numerous uses, such as predicting a likely result, supporting the analysis of problems, and aiding in making decisions. They have the advantage of being theoretically pure and have been shown to perform well on glitches.

In conclusion of this local literature, the ability of teachers to use Information and Communications Technology (ICT) to build a blended learning mode of delivery is critical. Various results were found regarding the university teachers' ICT skills, which demanded to send them to training and seminars on ICT technologies for learning and technology operations and concepts. Furthermore, a methodology for determining the instructors' ICT competencies was developed, with highly acceptable outcomes as evidenced by the decision tree, confusion matrix, Receiving Operating Characteristics Curve (ROC), and Area under the ROC Curve.

The above literature discussed a methodology for assessing the skills of instructors was also developed with highly acceptable outcomes as evidenced by the decision tree. The outputs of the evaluation were highly satisfactory. Therefore, the Decision tree is a reliable tool for prediction.

As stated by A. Dela Cruz (2018). The following are some of the reasons for choosing decision trees as the study's primary method: First, CHAID decision tree model is a simple, easy-to-understand, and interpretable model. Second, because it is quick to develop predictive algorithms, it is extremely suited and flexible for future data changes due to its short training time. Furthermore, the decision tree technique has great prediction accuracy in various sectors, making it a desirable and trustworthy option for this type of project. Treatment category relapse patterns are revealed with the use of this technology, and a mainly accurate predictor for treatment priority prediction is achieved.

In conclusion of this local literature, the Department of Health is concerned about predicting treatment outcomes in tuberculosis patients. Data mining can be utilized to make decisions in the health-care system. The decision tree based on the divide and conquer strategy is the most widely used for data exploration. The goals of this study are to develop a predictive data mining model for TB patients to locate relapse treatments and to define the elements that influence relapse treatments so that TB patients who are at risk can receive assistance, guidance, and appropriate warnings.

The above literature discussed that the decision tree model is a simple and interpretable method that can be used for various types of studies. Its quick development time and flexible nature make it a good choice for various types of projects.

As eloquently stated by R. Sonza (2020). The Decision Tree Algorithm is used to automate the suggestion for GAD solutions. Two entities, decision nodes and leaves, can be used to explain the tree. The decisions or final outcomes are represented by the leaves. The researcher is successful in identifying processes for gender and development in dynamic data sources. The use of the Decision Tree Algorithm to computer applications greatly aids the GAD Focal Point approach in identifying solutions to problems discovered through data collection. They have quick data sources that can help manage and monitor GAD projects, and they can identify a variety of GAD solutions that are required for planning, programming, and policy formulation.

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The above literature discussed the process of identifying gender and development processes in dynamic data sources. Through the use of the Decision Tree algorithm, the researcher can easily identify and implement solutions to various GAD problems. The Decision Tree Algorithm is a powerful tool that helps in identifying and implementing GAD solutions. It can be used to explain a tree, or it can be used to represent decisions or final outcomes.

Foreign Literature

The focus of this part is to take the relevant foreign literature as well as to better understand the correlation between technologies that is focused on this study.

According to M. Slater et. al (2016), physical protection, preservation, and restoration of sites are the best ways to safeguard cultural heritage. There has also been a lot of work done over the years on digital capture and visualization of such places, which can be exhibited in virtual reality. Furthermore, the first and most obvious use of VR in this field is to allow individuals all over the world to virtually visit and explore such sites. Except for the nature of the sight viewed, this is no different from virtual travel or tourism. This is also achievable at museums with virtual reality installations. The second is the digitalization of places for future generations, particularly those in danger of being destroyed due to factors such as climate change or violence. The third sort of application demonstrates how these locations would have appeared if they had been fully restored in the past and under various situations such as lighting.

In addition, it is probable that users will not need to physically visit a location to claim that they have visited it if they use virtual reality. Users could be navigating the streets and shopping in Hong Kong, climbing Mount Everest, seeing the Taj Mahal, touring the Forbidden City in Beijing, or even exploring the surface of Mars from the comfort of the users' own home. From Polynesia to Greenland, they may witness ceremonies and customs firsthand. This is a well-known and long-debated application. There are numerous possibilities: visiting a location virtually before going there, visiting the location instead of going there, having a business meeting virtually with remote partners, meeting in a shared virtual environment, taking a break on a beach in the middle of the day in the middle of the winter during your coffee break in the office; the possibilities are only limited by imagination and what technology can deliver which of course is always changing.

Moreover, people in the travel industry were already thinking about the "virtual threat to travel and tourism" two decades ago (Cheong, 1995), stating that "the imagined threat of virtual reality becoming a substitute for travel is not unwarranted and should not be ignored." Virtual reality has a number of significant

advantages over visiting a tourist attraction in person which could lead to virtual reality eventually replacing travel and tourism.” However, Cheong (1995) goes on to explain why this isn't necessarily a threat: virtual immersion isn't the same as actually being there; it would be difficult in VR to engage in exchanges with the locals (for example, discussions in a market, learning to dance the Hula); the real world has a level of complexity and randomness that can't be replicated in VR; people might confuse real and virtual reality.

In addition to this, of course, tourism has not been replaced by VR since 1995 but none of the issues raised above are insurmountable (even revenue from tourism could be protected by some kind of royalty system). Furthermore, as global warming becomes a more significant potential and hazard, virtual reality (VR) may offer a solution to mitigate some of the harmful effects of travel. According to Guttentag (2010), virtual reality (VR) could be valuable in tourism for planning, management, marketing, entertainment, and education, as well as allowing access to inaccessible regions like archeological sites and preserving heritage. Guttentag, on the other hand, questioned whether virtual reality could ever be a viable substitute for real travel, emphasizing a point made by Cheong (1995) that VR may never be able to replace basic sensory experiences such as "the smell of ocean spray" or make virtual surfing feel as authentic as the real thing. To put it another way, can virtual reality ever be technically capable of delivering a genuine substitute for the actual thing? However, this article by M. Slater et al. (2016) stated that “we do not attempt to answer this question, since the answer cannot be known”. This article also concludes that they primarily evaluated developments in virtual reality (VR) since its inception in the 1980s, with a focus on applications, particularly those having research-backed outcomes. The industry is continuously changing, and people's imagination is astounding, with new ideas and projects appearing on a regular basis. They have compiled a list of some recent concepts that have piqued their interest (as of May 2016). Typically, they are work-in-progress ideas with no outcomes or, in some cases, no level of implementation.

The above literature discussed that Virtual reality (VR) can be used to preserve cultural heritage sites. Virtual travel and tourism is no different from virtual travel or tourism, except for the nature of the sight

viewed. VR also allows people to virtually visit and explore such sites. This is also achievable at museums with virtual reality installations. The literature also discussed that the possibilities are only limited by imagination and what technology can deliver which is always changing. Moreover, Virtual reality (VR) may offer a solution to mitigate some of the harmful effects of travel.

According to the next literature, Head-Mounted Displays (HMD), such as Virtual Reality, are becoming increasingly useful for a wider range of construction and engineering applications, while also costing less than in the past (Hilfert, T., & König, M. 2016). Even non-programmers can create logic procedures and levels for presentation using modern gaming engines like Unreal Engine 4. Instead of needing to digest a complex explanation or offline produced 2D/3D pictures, Virtual Environments (VE) allow users to experience complex models or control schemes.

Furthermore, the article demonstrated the viability of automating significant aspects of the VR creation process using Building Information Modeling (BIM) as a starting point, as well as the simplification achieved by employing the proposed methodology and tools, as there is no need for a parallel design process for visualization.

When implementing the proposed workflow and plugin for UE4, BIMServer as a central server for storing BIM data is advantageous. Geometry can be imported rather than having to be recreated. In the future, users want to expand the capabilities and streamline the workflow to offer an easy importer for VR experiences based on real-world building data.

Evacuation plan testing, expert training, and environment accessibility validation are some of the other applications for using a VE. The scenarios provided are just a sample of what can be done with a VE for building and engineering. Users may find it easier to interact with virtual environments when adopting intuitive and natural control schemes.

In conclusion of this literature, the fact that graphics card makers are changing their designs and drivers to make them more VR-friendly appears to be a sign that this technology is here to stay and will only get better in the future.

The above literature discussed that Virtual Environments (VE) or Virtual Reality (VR) allow users to experience complex models or control schemes. A VE can be used for building and engineering. The scenarios provided are just a sample of what can be done with a VE in the real world. Users may find it easier to interact with virtual environments when adopting intuitive and natural control schemes.

According to the next literature, in this literature, students role-played in their target settings using Google Cardboard and Google Expeditions in an advanced Chinese Language class. A nurturing learning experience, according to Situated Cognition Theory, should help construct and integrate activity, context, and culture, resulting in rich implicit knowledge, learning, and acting (Xie, Y. et al., 2019). The VR technologies used in this study helped participants enhance their oral presentations by allowing them to engage in a process that included contextualized authentic learning, teacher coaching, peer cooperation, articulation of the information gathered, and reflection. Furthermore, the research was significant in several aspects for the field: For starters, study demonstrated that using VR technologies improved participants' speech ability in terms of content and vocabulary. Second, the qualitative analyses that followed provided insight into why learning had improved.

In conclusion of this literature, since mobile-based VR technologies are becoming more widely used in classrooms, in-depth evaluations of how students interact with them are critical for future VR design and research. Understanding the boundaries and repercussions of these tools' use in higher education has major relevance for instructors, practitioners, and instructional designers for the classroom as a new and potentially powerful technology. The research could also help to establish best practices for energizing language classrooms and outreach with the increasingly common mobile-based, head-mounted VR technology for meaningful language learning and growth.

The above literature discussed that using VR technologies improved participants' speech and vocabulary. The research was significant in several aspects for the field of virtual reality education, including how it helped students learn more effectively. Researchers say their findings could be used to improve oral presentations.

According to the next literature, the DE-ADT/SPV approach is introduced in this study, which uses a global search strategy to locate near-optimal axis-parallel Decision Trees (DT) represented using real-valued chromosomes. (R. Rivera-Lopez and J. Canul-Reich, 2017) This method uses a scheme to map possible DTs from the dataset whose model is built and estimates the size of the chromosomes depending on the dataset's properties. The DE-ADT-R/SPV method is an efficient DTI process, according to the experimental data, because it produces DTs with excellent precision and a lesser number of leaf nodes. The refinement of the best DT in the final population allows the model's training accuracy to be improved. Despite the fact that the DE-ADT/SPV technique does not generate superior results than the Random Forest (RF) algorithm and the Multilayer Perceptron (MLP)-based classifier, they are statistically equivalent.

Furthermore, the DE-ADT-R/SPV variation has the advantage of producing models with easily understandable decisions and operations; while the RF technique produces DTs as well, its voting process makes tracing the model's decisions extremely challenging. When a rule to map a DT from a real-valued chromosome is implemented, the Differential Evolution (DE) method is an effective approach for creating axis-parallel DTs. The DE operators may be employed without modification, and the chromosomes in the population reflect only possible DTs, which is a benefit of this strategy. This study concludes that a Decision Tree generated using this method can be used as a predictive model in fields such as biology, medicine, and finance, among others, where the model's interpretability has a crucial importance.

The above literature discussed that the DE-ADT/SPV approach uses a global search strategy to locate near-optimal axis-parallel Decision Trees (DT) represented using real-valued chromosomes. Study concludes that a Decision Tree generated using this method can be used as a predictive model in fields such as biology, medicine, and finance. However, this does not conclude that Decision Trees cannot be used in other aspects like, in the researchers case, in tourism.

According to the next literature, Binary Decision Tree Classifier based on K-means Splitting (BDTKS) is a new linear multivariate decision tree that is developed in this paper. The suggested methodology includes crucial components such as introducing K-means clustering as a node splitting mechanism and providing a non-split condition (Wang, F. et al., 2020). The former gives BDTKS a high degree of generalization, which improves classification performance. The latter allows the tree model to tolerate class imbalance cases more flexibly than MDT1 and 505 MDT2's majority class proportion based split criteria. As a result, BDTKS can adjust to either class balance or class imbalance situations.

BDTKS matches or beats the classic univariate decision trees (C4.5 and CART) and the latest multivariate decision trees (MDT1 and MDT2) in classification performance, according to experiments on 19 publicly available data sets. In terms of efficiency, BDTKS and its accelerated counterpart A-BDTKS require the same 510 training time as earlier decision trees but take less time to classify. As a result, in the proposed linear multivariate decision tree, superior classification performance and high efficiency coexist. When dealing with high-dimensional instances, however, the K-means node splitting approach in BDTKS is likely to take a long time.

Furthermore, while the proposed non-split condition in BDTKS has reduced the parameter's 515 sensitivity compared to the majority class proportion based split condition, there is still space for improvement. At each non-leaf node, feature selection or dimensionality reduction techniques can be investigated in future work. To further reduce parameter sensitivity, the non-split condition might be advanced. The suggested BDTKS decision tree 520 can be ensembled to create decision forests, resulting in even better performance.

The above literature discussed that Binary Decision Tree Classifier based on K-means Splitting (BDTKS) is a new linear multivariate decision tree that is developed in this paper. The suggested methodology includes crucial components such as introducing K-means clustering as a node splitting mechanism and providing a non-split condition. The former gives BDTKS a high degree of generalization, which improves classification performance. As a result, BDTKS can adjust to either class balance or class imbalance situations. BDTKS matches or beats the classic univariate decision trees (C4.5 and CART) and the latest multivariate decision trees (MDT1 and MDT2) in classification performance, according to experiments on 19 publicly available data sets.

Lastly, according to Packt Publishing (2018), a Binary Decision Tree is a structure that uses a sequential decision procedure to make decisions. A feature is examined from the root, and one of the two branches is chosen. This process is repeated until the final leaf is reached, which is usually the categorization target that the users are seeking for. In addition, In comparison to other algorithms, Decision Trees appear to have simpler dynamics. However, if the dataset can be separated while maintaining internal balance, the whole procedure is simple and quick to forecast. Furthermore, because the fundamental structure of Decision Trees is unaffected by the values taken by each feature, they can perform effectively with unnormalized datasets.

The above literature discussed that a Binary Decision Tree is a structure that uses a sequential decision procedure to make decisions. The fundamental structure of Decision Trees is unaffected by the values taken by each feature. This means they can perform effectively with unnormalized datasets, according to Packt Publishing (2018)

According to the researchers, this is the main reason why the Binary Decision Tree algorithm is the chosen algorithm to fit in creating the CCC Tour Guide. Due to its simplicity, being straightforward and uncomplicated, Binary Decision Tree algorithm was chosen.

Related Studies

Local Studies

The focus of this part is to take the relevant local literature as well as to better understand the correlation between technologies that is focused on this study.

According to M. A. Timbal (2019), a decision tree is a flowchart with a tree structure, in which the internal nodes represent attribute validations, the branches represent validation outputs, and the leaf nodes represent classes. The root node is the node in the highest part of the tree. When you reach a "Leaf node," the instance is classified according to the class assigned by that node. The flow of the tree from the root down, according to the values that have the attributes for each node, and when you reach a "Leaf node," the instance is classified according to the class assigned by that node.

In conclusion of this local study, a decision tree-based classification technique is employed to build a rule-based classifier. According to the findings, three extracted rules from the decision tree produced using the partykit and rpart libraries as a predictive model, a basis for predicting who among the registered students will dropout, were identified.

The above study discussed a local study that used a decision tree-based classification technique to build a rule-based classifier. Three extracted rules from the decision tree produced using the partykit and rpart libraries as a predictive model, a basis for predicting who among the registered students will dropout, were identified.

As stated by M. Gotardo (2019). The characteristics of the Data Structures and Algorithms course that will affect students' performance were the focus of this research. The decision tree model was created using the J48 method, which revealed that the Finals characteristic had the greatest indication, which is critical for students passing the subject. Most importantly, a model for determining students' performance in data structures and algorithms was developed, as evidenced by the Decision Tree, Confusion Matrix, ROC,

and Area under the ROC Curve. Furthermore, institutions can use the data-driven system to track student success.

In conclusion of this local study, educators can collect feedback on student learning experiences in face-to-face encounters with their students in traditional classroom environments, allowing for continuous evaluation of their teaching programs. However, with the integration of technology in learning contexts, educators must discover additional ways to obtain this information. The prediction model's findings will aid educators in taking steps to improve students' performance.

The above study discussed that a new model for determining students' performance in data structures and algorithms was developed: Decision tree. The decision tree model was created using the J48 method, which revealed that the Finals characteristic had the greatest indication. The prediction model's findings will aid educators in taking steps to improve student's performance. Therefore, Decision Tree is a good choice for predictive analysis.

As eloquently stated by A. Claridades (2018). The photos were combined to create the 3D virtual tour using PanoTour, based on how each Shooting Point is connected to an adjacent Shooting Point. The position of adjacent Shooting Places would be highlighted in PanoTour, and the accompanying view from those points would be indicated in the associated photos. This creates a virtual tour that can be seen in any web browser and provides an immersive and seamless representation of navigation in a 3D indoor setting.

In conclusion of this local study, this visualization was created by connecting photos so that the findings may be combined with 3D spatial dataset standards like IndoorGML, which describe relationships between spaces in an image. Because humans perceive locations better visually and easily in 3D rather than in terms of textual directions, the output could be used as a navigational aid throughout the building interior. This also cuts down on the amount of abstraction needed to create representations because objects and scenes appear exactly as they do in the photos.

The above literature discussed that the photos were combined to create the 3D virtual tour using PanoTour, based on how each Shooting Point is connected to an adjacent Shooting Point. This creates a virtual tour that can be seen in any web browser and provides an immersive representation of navigation in a 3D indoor setting. Since users perceive locations better visually and easily in 3D rather than in terms of textual directions, the output could be used as a navigational aid throughout the building interior.

As mentioned by B. Grepon (2021). The project employed virtual reality (VR) technology to observe not only the surface but also the interior structure of an architectural design, allowing participants to move around in the virtual world, see it from different angles, reach into it, grab it, and alter it. It aims to create an Android application as an alternative method of promoting the college and its campus through Architectural Visualization using Virtual Reality to tour prospective students remotely and allows users to view the current and actual architectural structure of the school buildings without having to be physically present in the area, which can be navigated using a smartphone that runs an android operating system.

In conclusion of this local study, Virtual Reality Platform allows students from various secondary schools located near the college to experience the impression of being within the institution without having to physically enter. Using VR Gadgets to navigate buildings is still relatively new in the community, making the VR app a hit with those who use it. With the advancement of technology, virtual reality (VR) plays an important role in allowing people to see what's inside a building and navigate about it without actually being there.

The above literature discussed that Virtual reality (VR) allows people to see what's inside a building and navigate about it without having to be physically present in the area, which can be navigated using a smartphone that runs an android operating system. Using VR is still relatively new in the community, making the VR app a hit with those who use it.

As studied by S. Urizza et. al (2016). With a virtual reality tour, potential visitors may now experience these hidden natural wonders without really being there. Uncertainties about what to expect in the area would

be reduced, allowing them to explore and roam around inside, leading to the formation of the Panguil River Eco-3D Park's Mobile Virtual Tour Application (MoViT), which was created and designed using story conception, concept scripting, feasibility analysis, script writing, system evaluation, revision, and integration. The study was able to deliver an eco-park experience to potential visitors, giving them a "feel" for the park.

In conclusion of this local study, virtual tours will allow visitors and potential visitors to experience being within the Panguil River Eco-Park in a virtual environment. The study was based on usability testing results, which were divided into four categories: visual aesthetics, interactivity, sensory simulation, and navigation. Even though the users did not hike to the falls, they were able to get a peek of the eco-park and the waterfalls, but they also mentioned that the sound of the program might be improved to make it louder. In terms of potential tourists, they have stated that despite having little knowledge of the eco-park, they were able to travel about and get a "feel" for it via the mobile virtual tour app.

The above literature discussed that The Panguil River Eco-3D Park's Mobile Virtual Tour Application (MoViT) was able to deliver an eco-park experience to potential visitors, giving them a "feel" for the park. Even though the users did not hike to the falls, they were able to get a peek of the eco-Park and the waterfalls.

Foreign Studies

The focus of this part is to take the relevant foreign studies as well as to better understand the correlation between technologies that is focused on this study.

According to Bogusevschi, D. et al. (2020), Virtual Reality (VR) is a technology with a lot of potential and a lot of pedagogical benefits that can be used to enable new teaching approaches. VR creates an immersive multimedia 3D representation of real life that encourages interaction with the produced world and allows for sensory experiences. Virtual reality offers a wide range of applications in a variety of fields, including physics.

In addition, this paper builds on the utilization of the NEWTON Project 3D immersive Water Cycle in Nature application as part of a case study from the learner experience point of view (Bogusevschi, et al., 2018). The Virtual Reality (VR) and Virtual Laboratory technology (VL) application, which is computer-based, focuses on physics terms related to the natural water cycle and precipitation creation, such as vaporisation, evaporation, boiling, and condensation.

Digging deeper in this study, The Water Cycle in Nature computer-based VR - VL application for teaching Physic concepts is described in this publication, which is part of the Horizon 2020 NEWTON Project. There is a description of the application as well as its educational material, which focuses on natural events that contribute to precipitation production. The usefulness of the VR - VL simulation program as a learning environment and its effects on user experience were explored in a case study conducted in a secondary school. A Learner Satisfaction Questionnaire was utilized to get feedback on the application and technology deployed. In addition to the previously mentioned statistically significant knowledge gain achieved by the experimental group when using the application versus the control group (Bogusevschi, et al., 2018), it was also shown to provide students with a positive learning experience, as over 74 percent of students enjoyed learning using the Water Cycle in Nature application.

In conclusion of this study, Virtual reality (VR) creates an immersive multimedia 3D representation of real life that encourages interaction with the produced world and allows for sensory experiences. The Water Cycle in Nature computer-based VR - VL application for teaching Physics concepts is described in this publication, which is part of the Horizon 2020 NEWTON Project. It was shown to provide students with a positive learning experience, as over 74 percent of students enjoyed learning using the Virtual Reality in Nature application.

The above study discussed that Virtual reality (VR) creates an immersive multimedia 3D representation of real life that encourages interaction with the produced world and allows for sensory experiences. The Water Cycle in Nature computer-based VR - VL application for teaching Physics concepts is described in this publication, which is part of the Horizon 2020 NEWTON Project. It was shown to provide students with a positive learning experience, as over 74 percent of students enjoyed learning using the Virtual Reality in Nature application. A Learner Satisfaction Questionnaire was utilized to get feedback on the application and technology deployed.

Proceeding to the next study, a decision tree is a flowchart-like tree structure in which each internal node represents a test on an attribute, each branch represents a test outcome, and each leaf node (or terminal node) represents a class label (Sharma, H., & Kumar, S., 2016). The attribute values of a tuple X are evaluated against the decision tree given a tuple X . A path is traced from the root to a leaf node that contains the tuple's class prediction. Decision trees are simple to translate into categorization rules. A decision tree is used as a predictive model in decision tree learning. It translates observations about an object to conclusions about the item's goal value. In statistics, data mining, and machine learning, it is one of the predictive modeling methodologies. Classification trees are tree models in which the goal variable can take a finite set of values. In this tree structure, leaves indicate class labels and branches represent feature combinations that lead to those class labels. Furthermore, when compared to other categorization approaches, constructing a decision tree is rather quick. Trees can be used to create SQL statements that

can be used to effectively access databases. When compared to other classification approaches, decision tree classifiers achieve similar or superior accuracy.

In conclusion of this study if applied on the researchers' application, a decision tree is a structure that shows how an attribute is evaluated. Each internal node has a test, and each branch has a result, and each leaf node has a class label. If applied on the researchers' application, each node can be set to rooms and each branch can be set to path ways, and each leaf node's class label can be set to the categories of the room. (e.g. club rooms, classrooms, etc.)

Furthermore, the path to the next leaf node is traced from the root. A decision tree is a simple to implement framework that translates an object's observations into rules. A classification tree is a tree model that takes a set of values and shows the feature combinations that lead to a class label. This structure allows for quick construction.

The above study discussed that a decision tree is a flowchart-like tree structure in which each internal node represents a test on an attribute, each branch represents the test outcome, and each leaf node (or terminal node) represents a class label. Decision trees are used as a predictive model in decision tree learning. When compared to other classification approaches, decision tree classifiers achieve similar or superior accuracy. Also, the decision tree is a simple to implement framework that translates an object's observations into rules. This structure allows for quick construction and easy to implement building up.

According to the next study, as stated by Bombara, G., et al. (2016), "A decision tree is a tree-structured sequence of questions about the data used to make predictions about the data's labels." Also, the root is the initial node in a tree; the depth of a node is the length of the path from the root to that node; the parent of a node as the neighbor whose depth is one less; the children of a node as the neighbors whose depths are one more. A leaf is a node that has no children; all other nodes are considered non-terminal nodes. The researchers focus on binary decision trees in this study, where each non-terminal node divides the input into two children nodes and each leaf node predicts a label.

Furthermore, the proposed framework covers decision-tree learning algorithms that can be customized by supplying three elements: (a) a selection of primitive attributes of interest; (b) an impurity measure that captures the node's homogeneity; and (c) algorithm halting circumstances.

The above study discussed that a decision tree is a tree-structured sequence of questions used to make predictions about the data's labels. The researchers focus on binary decision trees in this study, where each non-terminal node divides the input into two children nodes and each leaf node predicts a label. The proposed framework covers decision-tree learning algorithms that can be customized by supplying three elements: primitive attributes, impurity measure and halting circumstances.

According to the next study, "Experiments in Induction," the first documented study to describe the nature of decision trees, was published in 1966. The decision tree is a hierarchical consistent form for displaying rules. If the tree is binary, the primary component of this structure is the answer "Yes" or "No," or a series of questions defined dynamically when trees have an arbitrary number of branches. (N. Pashynska, et al., 2016)

The model, which is presented as a decision tree, is intuitive and favorable to comprehending the problem at hand. This characteristic of decision trees is valuable in evaluating the categorization model as a whole, not just in referring a new object to a class. The decision tree makes it possible to understand and explain why a particular object belongs to a particular class. Moreover, in situations where it is difficult for analytics to formalize information, decision trees can develop categorization models. The user is not required to select input attributes while using the decision tree algorithm (independent variables). All existing traits can be included in the algorithm's input; the algorithm will choose the most essential among them, and only those will be utilized to build the tree.

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for analytics to formalize information, decision trees can develop categorization models. Also, it is said that the user is not required to select input attributes while using the decision tree algorithm

According to the next study, The Unity3D-based 3D virtual tour guide training system provides trainers with a teaching environment that is rich in interactivity and immersion, allowing them to "travel to" the beautiful site to study and practice anytime and anywhere (Wang, L., 2019). The use of this system can not only effectively solve the time, cost, traffic safety, and other issues that currently exist in guide training, but it can also be combined with classroom teaching, complementing each other's advantages, and attempting to reform and innovate teaching modes in order to improve teaching quality.

The above study discussed that the Unity3D-based 3D virtual tour guide training system provides trainers with a teaching environment that is rich in interactivity and immersion, allowing them to "travel to" the beautiful site to study and practice anytime and anywhere.

Technical Background

From 1970 through 1990, the virtual reality industry primarily produced VR systems for medical, flight simulation, vehicle design, and military training. But in the mid-1990s, Virtual reality headsets for consumers were first introduced by video game firms (The Board of Trustees of the University of Illinois, 1995). Oculus (Rift), HTC (Vive), and Sony (PlayStation VR) all introduced next-generation commercial tethered headsets in the 2010s, igniting a new wave of application development (F. Rohr, 2015). Nowadays, Virtual Reality is often used for tour guiding (e.g. Google Maps) and travelling purposes. Such applications are helpful for people who do not travel that much and are not familiar with many places.

In this study, the proponents used Google Street View to develop the 360 Photo Spheres that are being used in the application and an android mobile device to capture the 360 Photo Spheres from the institution.

The Waterfall model of System Development Life Cycle is used to make certain that programs were effectively completed step-by-step. This implies that each stage of the development process can only start after the one before it has finished.

The Binary Decision Tree algorithm will be used in the study to virtually structure the institution which will help in guiding the users. The application used C# as the programming language. The application was developed using Unity 3D to virtually modify the virtual elements that were used in the system. Unity 3D was also used to code and implement the algorithm. For the compilation of the created sphere photos, Unity 3D is being used.

Synthesis

A literature discussed that Virtual tour's spatial presence increased participants' interest in universities and their desire to visit campuses. Virtual tours were also an effective tool for informing tourists about universities. They also discussed that using VR technologies can improve participants' speech and vocabulary. It was significant in several aspects for the field of virtual reality education, including how it helped students learn more effectively. Some researchers say their findings could be used to improve oral presentations.

If applied to this study, a virtual tour can be applied to City College of Calamba to increase new students' interest and their desire to visit the institution. It can also be said that it can improve new students' knowledge about the institution.

Another study discussed that Virtual reality (VR) allows people to see what is inside a building and navigate about it without having to be physically present in the area, which can be navigated using a smartphone that runs an android operating system. Furthermore, some study discussed that Virtual Tour Application, like MoViT, was able to deliver an eco-park experience to potential visitors, giving them a "feel" for the park.

If applied to this study, it can be said that the researchers' application, CCC Tour Guide, can allow people to see what is inside the building and navigate about it without having to be physically present in the area, which can be navigated using a desktop that runs a Windows operating system. It can also be said that the researchers' application can be able to deliver a City College of Calamba experience to potential visitors to try to give them the "feel" of the institution.

Furthermore, another literature discussed that Decision trees were effective as a tool for prediction. They discussed a methodology for assessing the skills of instructors that was also developed with highly acceptable outcomes as evidenced by the decision tree. The outputs of the evaluation were highly satisfactory. It can also be said that the decision tree model is a simple and interpretable method that can be

used for various types of studies. Its quick development time and flexible nature make it a good choice for various types of projects.

If applied to this study, Decision Trees can be an effective tool to predict new students' desire to travel around the institution in terms of directions.

Moreover, the Decision Tree Algorithm is a powerful tool that helps in identifying and implementing Generalized anxiety disorder (GAD) solutions which can also be applied in different scenarios. It also discussed that the DE-ADT/SPV approach uses a global search strategy to locate near-optimal axis-parallel Decision Trees (DT) represented using real-valued chromosomes. The study concludes that a Decision Tree generated using this method can be used as a predictive model in fields such as biology, medicine, finance. However, this does not conclude that Decision Trees cannot be used in other aspects like, in the researchers' case, in tourism.

Another literature discussed that Binary Decision Tree Classifier based on K-means Splitting (BDTKS) is a new linear multivariate decision tree. The suggested methodology includes crucial components such as introducing K-means clustering as a node splitting mechanism and providing a non-split condition. The former gives BDTKS a high degree of generalization, which improves classification performance. If applied in this study, Binary Decision Tree Classifier can be used to improve classification performance based on the institution's architectural design.

The studies discussed a local study that used a decision tree-based classification technique to build a rule-based classifier. Three extracted rules from the decision tree produced using the partykit and rpart libraries as a predictive model, a basis for predicting who among the registered students will dropout, were identified.

If applied in this study, it can be said that decision tree-based classification technique can be used as a predictive model for predicting where the students' or visitors' will likely to be toured at.

Furthermore, a study discussed that a new model for determining students' performance in data structures and algorithms was developed: Decision tree. The decision tree model was created using the J48 method, which revealed that the Finals characteristic had the greatest indication. The prediction model's findings will aid educators in taking steps to improve student's performance. Therefore, Decision Tree is a good choice for predictive analysis. If applied in this study, it can be said that Decision Tree is a good choice for predictive analysis since it can be used on different aspects.

Some studies discussed that a decision tree is a flowchart-like tree structure in which each internal node represents a test on an attribute, each branch represents the test outcome, and each leaf node (or terminal node) represents a class label. Furthermore, Decision trees are used as a predictive model in decision tree learning. When compared to other classification algorithms, decision tree classifiers achieve similar or superior accuracy. Also, the decision tree is a simple to implement framework that translates an object's observations into rules. This structure allows for quick construction and easy to implement building up.

If applied on the application, each node can be set to rooms and each branch can be set to path ways, and each leaf node's class label can be set to the classification or directions of the room (e.g. left, right, forward, backward). It can also be said that Decision trees allow quick construction and easy implementation, which in the researchers' case, is a good choice of algorithm.

Moreover, the decision tree is a hierarchical consistent form for displaying rules. If the tree is binary, the primary component of this structure is the answer "Yes" or "No" In situations where it is difficult for analytics to formalize information, decision trees can develop categorization models. Also, it is said that the user is not required to select input attributes while using the decision tree algorithm.

If applied in this study, the researchers can input all of the possibilities in terms of classification or direction. With this said, the algorithm can predict all of the possible directions that the user will likely to travel at.

Another study discussed that a Unity3D-based 3D virtual tour guide training system provides trainers with a teaching environment that is rich in interactivity and immersion. If applied in this study, the researchers' application, CCC Tour Guide, can provide new students and users a rich and interactive immersion to the institution.

In conclusion of these reviewed literature and studies, it can be said that Binary Decision Tree Algorithm is a good algorithm to be applied in this application since it can improve new students' knowledge about institutions, it can be able to deliver a City College of Calamba experience to potential visitors to try to give them the "feel" of the institution, it can be an effective tool to predict new students' desire to travel around the institution in terms of directions, and it can also be used as a predictive model in fields such as biology, medicine, finance, especially, tourism. Furthermore, it can be used to improve classification performance based on the institution's architectural design, as a predictive model for predicting where the students' or visitors' will likely to be toured at, and it is also a good choice for predictive analysis since it can be used on different aspects. Moreover, when compared to other classification algorithms, decision tree classifiers achieve similar or superior accuracy. Also, the decision tree is a simple to implement framework that translates an object's observations into rules. This structure allows for quick construction and easy to implement building up. Some study concluded that it is a good algorithm for predictive analysis.

CHAPTER III

DESIGN AND METHODOLOGY

This chapter focuses on the discussion of the research methods and procedures adhered to by the researchers in order to answer systematically the specific problems posed for investigation. This includes research design, algorithm design and techniques, software development methodology. This chapter presents a comprehensive analysis and design required in every system development phase.

Research Design

The research will look into the relationship between the variables. Specifically, the impact of Binary Decision Tree algorithm on the application. The study will describe the developmental stages of the program. The Developmental method of research will be implemented in this study. Since the study is experimental research, the researchers will employ the Waterfall model as the System Development Life Cycle. The data gathering will be conducted at the requirement analysis and specification phase. Since the researchers are required to conduct a survey to senior high school students, an online survey was conducted via Google Forms. Since an online survey was conducted and the data gathering was conducted at the requirement analysis and specification phase, the research design would be a mixed method of research. The researchers use the qualitative and the quantitative method approaches of research.

Algorithm Design and Techniques

Binary Decision Tree is the algorithm that has been implemented in the application of this study. In this study, the Decision Theory can be applied since this study will be using Binary Decision Tree Algorithm in which the users will pick and the outcome will be based on the user's picked choices in the program. The outcome will be the path that the Binary Decision Tree Algorithm will evaluate depending on their picked choices.

The application contains virtual elements that have the institutions' information. The Binary Decision Tree algorithm will be used to estimate the outcomes of choices based on user's selection. The study will also prove that the Binary Decision Tree Algorithm is the correct algorithm to implement in the program.

Binary Decision Tree algorithm is a structure that uses a sequential decision procedure to make decisions. A feature is examined from the root, and one of the two branches is chosen. The two branches technically asks a true or false questions. The process is repeated until the final leaf is reached, which is usually the categorization target or, in this case, the location that the user is seeking for. If applied in this application, the decisions are the choices of the user if they want to go left or right.

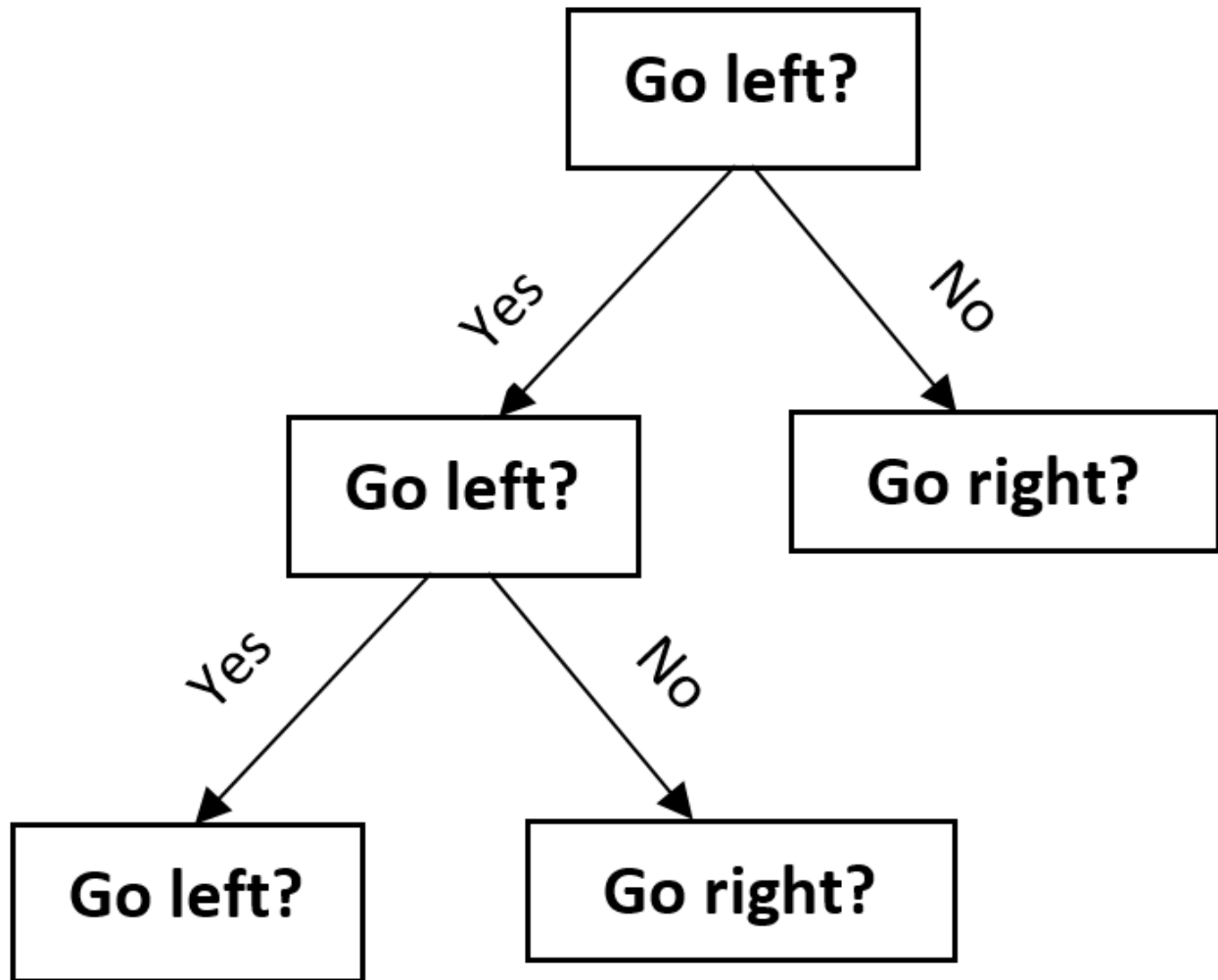


Figure 1: Binary Decision Tree Algorithm Technique

As shown on Figure 1, this is how the algorithm will be applied on the application. The algorithm can only be applied if there are at least two choices of path ways. The first text box from above is the parent node in this example then the below textboxes are its child. Technically, the algorithm asks if the user wants go left and the answer is only Yes, if the user wants to go left, or No, if the user wants to go right. The process is repeated until the final child is reached.

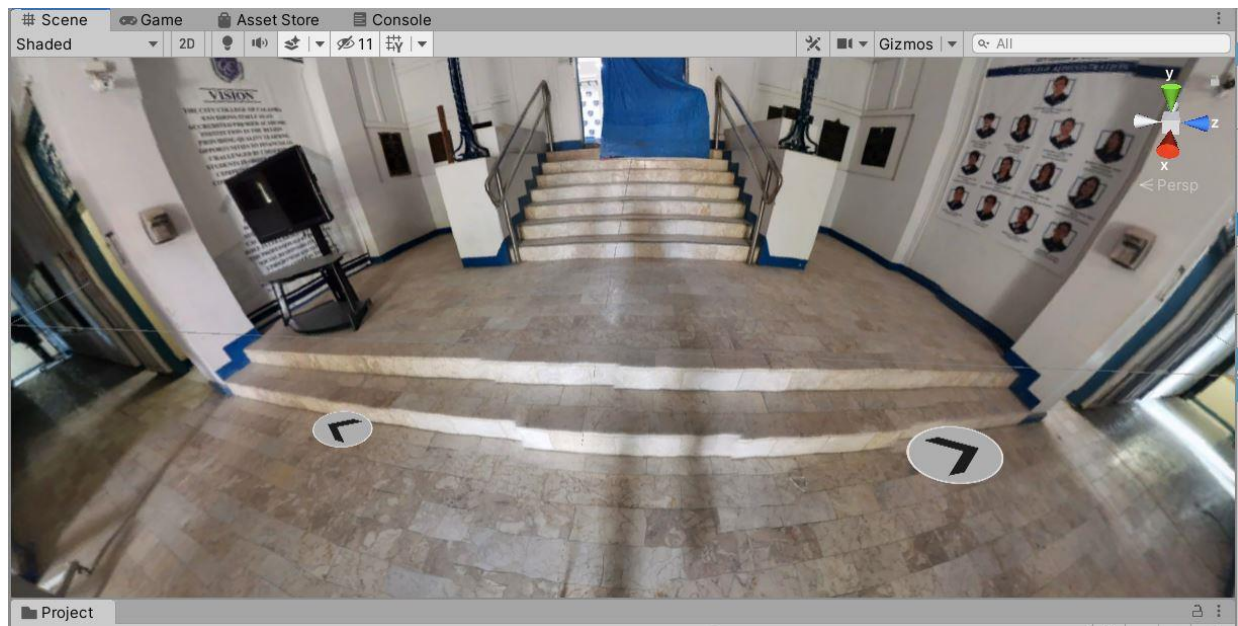


Figure 2: Sample of Application of the Algorithm

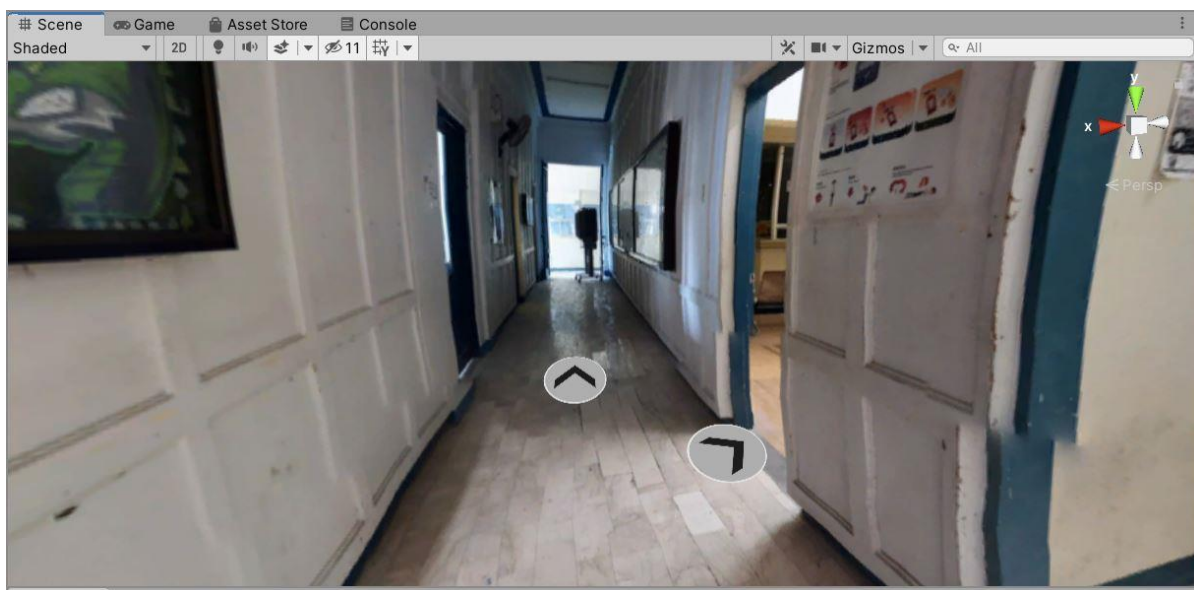


Figure 3: Sample of Application of the Algorithm

As shown on Figure 2 and Figure 3, this is what the algorithm will look like if applied in the application. In Figure 2, the algorithm technically asks if the user wants to go left or right. If the user wants to go left the user will click the arrow that points in that direction and vice versa. The process is repeated until the final

node is reached or until the application reached the satisfaction of the user because there are arrows to go back to further explore the institution. In Figure 3, this is the outcome if the user picked the left arrow on Figure 2. Another set of choices are shown depending on the architectural structure of the institution.

```

// first class
public void lobbyToLeft_btn()
{
    // 1st group
    canvasMainMenu.SetActive(false);
    view_cccmMainLobby.SetActive(false);
    lobby_toRight_CV.SetActive(false);
    lobby_toLeft_CV.SetActive(false);
    lobby_toBack_CV.SetActive(false);
    // 2nd group
    view_cccLeftHallway.SetActive(true);
    leftHallway_toForward_CV.SetActive(true);
    leftHallway_toRight_CV.SetActive(true);
    leftHallway_toBack_CV.SetActive(true);
}
// second class
public void lobbyToRight_btn()
{
    // 3rd group
    canvasMainMenu.SetActive(false);
    view_cccmMainLobby.SetActive(false);
    lobby_toRight_CV.SetActive(false);
    lobby_toLeft_CV.SetActive(false);
    lobby_toBack_CV.SetActive(false);
    // 4th group
    view_cccRightHallway.SetActive(true);
    righthallway_toForward_CV.SetActive(true);
    righthallway_toBack_CV.SetActive(true);
}

```

Figure 4: Sample Algorithm Code

As shown in Figure 4, this is an example of the algorithm that is applied on the application. This code comes from Figure 2 where the first choices were needed to be implemented. As shown in Figure 4, the first class is the button to go left and the second class is the button to go right. As shown on the first class, there are texts that is colored blue. These blue colored texts represents the choice of the users that was shown on Figure 1 which is the yes (true) or no (false). The code was grouped into four categories since they have different purposes. The purpose of the first group and third group is to hide the lobby. The purpose of the second group and fourth group is to display the left side of the hallway. The process is repeated until the final node is reached.

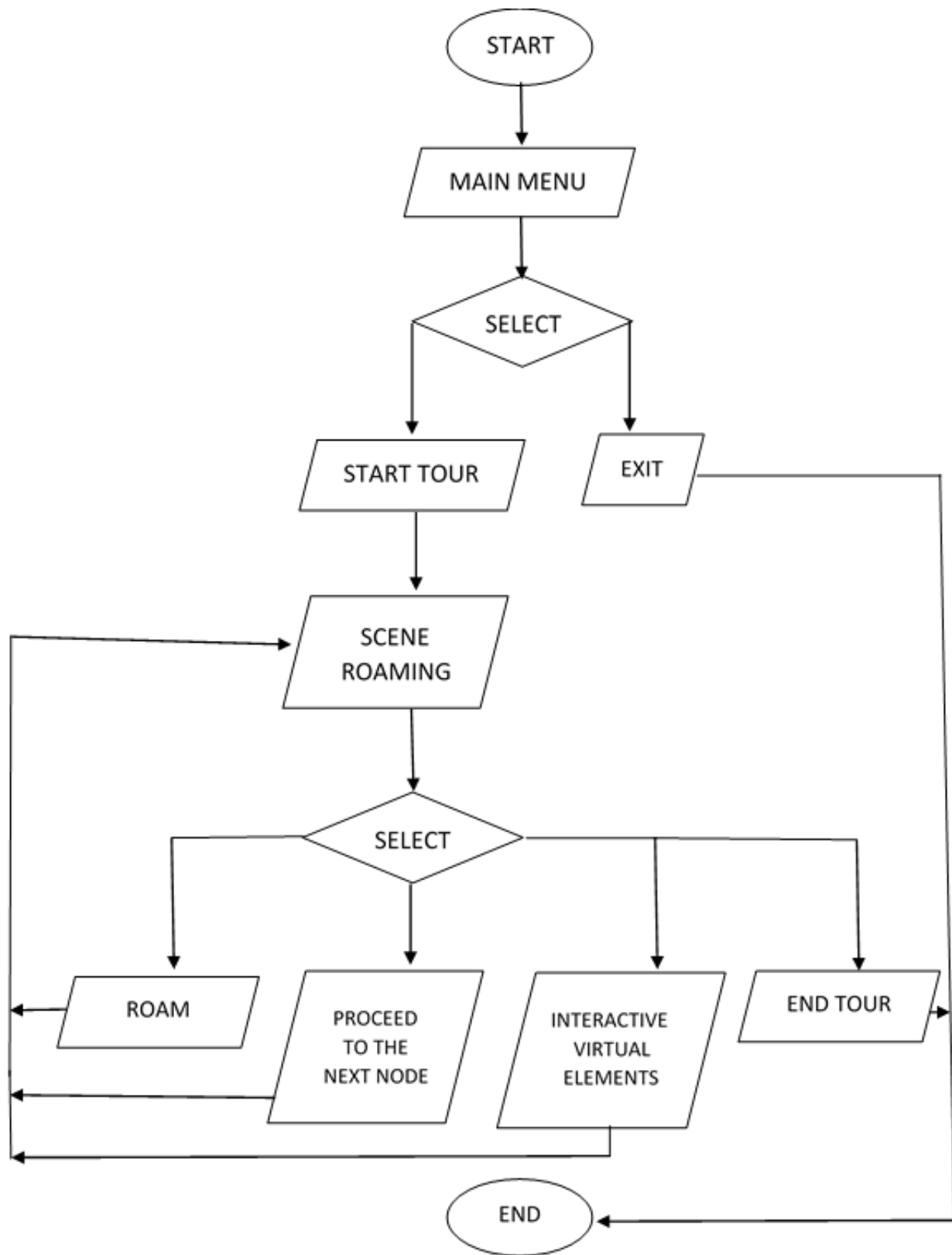


Figure 5: System Flowchart

As shown in Figure 5, this is the System Flowchart of the application. When the user starts the application, the program will display its main menu which has two choices. The first choice is to start the tour and the second choice is to exit the program. If the user picks the second choice, the program will immediately shutdown but if the user picks the first choice, the user will be directed to the first scene of the program. The user will then be able to roam the scene using the mouse. If the user clicks the mouse and drag, the user can be able to see the surroundings. If the user clicks the arrow, the user will be directed to the next scene or node. If the user clicks a virtual element, the user will be able to view and read the details about it. The controls are available and can be repeated if the user met the satisfaction of the tour and he or she can freely close the application.

Software Development Methodology

The study employed descriptive data presentations and Developmental data analyses using data from documentation and observation. The study uses the Binary Decision Tree algorithm to virtually structure the institution that will help to guide the users. C# is used as the programming language and Unity 3D as the programming platform. The researchers used Google Street View as it can produce Photo Sphere pictures from the institution.

The Waterfall Approach was the first SDLC Model to be widely used in Software Engineering to ensure that projects were completed successfully. The "Waterfall" technique divides the entire software development process into many sections. In this Waterfall approach, the output of one phase is typically used as the input for the next phase. The software development process is depicted as a series of events in the waterfall diagram. This means that each step of the development process can only begin once the preceding one has completed. In this lifecycle model, the phases do not overlap.

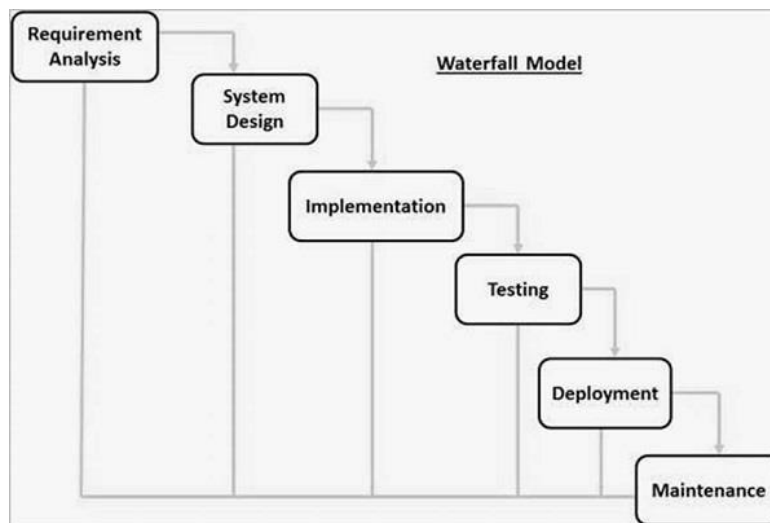


Figure 6: Waterfall Life Cycle Model

Requirement Analysis

This is the first phase of the waterfall lifecycle model. The proponents gathered and analyzed all requirement that can used as a basis for making the system reliable. This phase, Requirement Analysis, captures all potential requirements for the system to be constructed and documents them in a requirement specification document. This is where data are gathered and related studies about tour, algorithm and theories.

In this phase, the proponents conduct the various research and observation in order to have and better way of developing the system to came up the concept and designs for this study. The proponents gathered all the available resources then check if the resources are enough to design and develop the system. The proponents also conducted an online survey if the target users would like to try the application.

System Design

System Design researches and prepares the system design based on the needs specifications from the preceding phase. This system design aids in the creation of overall system architecture as well as the specification of hardware and system requirements.

In this phase, since the requirement analysis confirmed that the idea of the system is feasible, the proponents decided to propose the system to the panels. This is where the 360 Photo Spheres were captured. Moreover, the Photo Spheres helped the proponents to easily design and develop the system. The algorithm was also decided in this phase.

Implementation

The system is first built as discrete programs called units, which are then merged in the next phase, using inputs from the system design. Unit testing is the process of developing and testing each unit for its functioning.

In this phase, the 360 Photo Spheres were inputted into the program. The algorithm was also implemented in this phase. An example of unit that was developed in this phase is the script of the algorithm.

Integration and Testing

In this step, Integration and Testing, all of the units created in the implementation phase are merged into a system once they have been tested. After the system has been integrated, it is tested for any problems or malfunctions.

In this phase, the algorithm was checked thoroughly if it is the right algorithm for the system. All buttons and functions were checked thoroughly if working properly.

Deployment of System

In this step, System Deployment, the product is deployed in the customer environment or released into the market after functional and non-functional testing.

Maintenance

During this period of maintenance, there are a number of issues that develop in the client environment. To resolve these problems, updates have been released. Newer versions of the product have been created in order to improve it. Maintenance is performed in order to effect these changes in the customer's environment.

In this phase, the application will be finished after the institution finished its renovation.

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APPENDICES